

HOMEWORK 1.2

Installing PostgreSQL and Creating Databases

Installation Steps Summary:

1. Download the PostgreSQL installer for your OS from the official PostgreSQL website.
2. Launch installer and follow the setup instructions. Select default settings and create a password for the user.
3. To access the PostgreSQL server after installation, launch pgAdmin or the command line.
4. Name your database. Use an SQL command or pgAdmin to do this.
5. Connect to the database created.
6. Depending on what is to be stored, use SQL commands or pgAdmin to create a table with columns like ID, name, etc.
7. View the table in pgAdmin or list all tables to confirm that it was created. Go ahead with transacting the database.

Some Example Queries:

- UPDATE public."user" SET skills = '{"GitHub API",Gitlab,"Node.js","React.js","WebRTC"}'::text[] WHERE user_id = 3;
- SELECT * FROM public."user" ORDER BY user_id ASC
- DELETE FROM public."user" WHERE user_id IN (2);

Database and Tables:

	user_id [PK] integer	email_id character varying (255)	password_hashed character varying (255)	role character varying (50)	skills text[]	preferences text[]
1	1	lekhana@gmail.com	\$2b\$12\$92XGZeXj0XfscG4tZEtO.BosTuTVffjD6KnoEDsAY7RfiihD8Apq	manager	{}	{}
2	2	manaswini@gil.com	\$2b\$12\$IUZ07pvcISsVM1NnIUOateVz2OAnO/fV7fub96K1XhQmy1ynWoU...	manager	{}	{}
3	3	manvitha@gmail.com	\$2b\$12\$02GdXeSdMX6sn6RjTE.JweVgkctfSkjpc5tv2od.oDJQubFEFEW/W	employee	{'GitHub API',Gitlab,Node.js,React.js,WebRTC}	{Github}
4	4	nithya@gmail.com	\$2b\$12\$FGYh1qz8vlz.5BpCa4ggLgNjR.3AX49cnSdKp3Jz3lrIH99K1x9u	employee	{'Deep Learning','Machine Learning'}	{'Deep Learning'}
5	5	krupa@gmail.com	\$2b\$12\$EafoynP85/rf6zbaefbEkuHlmFFt0pRzYB0iCtB.plaJUWlmW5ye	employee	{React.js,Bootstrap,WebRTC}	{React.js}
6	6	manasa@gmail.com	\$2b\$12\$PIEwxYzPRoCgH60xV1WbEu01Jz9W8LCQ8M4DBqnZJdLtrCVHp...	employee	{'AST Parsing',Testing}	{'AST Parsing'}
7	11	lekdo30@gmail.com	\$2b\$12\$Y5nFj4/xm96T58yH3IQ5.ejH8JwgyMiMbbcfjEA7MzGRO2Ae2EM...	employee	{WebRTC,Github}	{WebRTC}

Benefits of PostgreSQL and pgAdmin:

- PostgreSQL is free and open source. It supports SQL querying and JSON. It is secure and works on Windows, Linux, and macOS.
- pgAdmin is used to simplify management of PostgreSQL. It has a simple, easy UI to create databases, tables and run queries. Unlike the command line. We can also back up and restore tables easily.